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Education

Institute of Science Tokyo (formerly Tokyo Institute of Technology)¹ <i>Doctor of Science in Mathematics</i> Supervisor: Kazuma Shimomoto	Tokyo, Japan <i>Apr 2024–Mar 2026</i>
Tokyo Institute of Technology <i>Master of Science in Mathematics</i> Supervisor: Kazuma Shimomoto (2nd year), Fumiharu Kato (1st year)	Tokyo, Japan <i>Apr 2022–Mar 2024</i>
Tokyo Institute of Technology <i>Bachelor of Science in Mathematics</i> Supervisor: Fumiharu Kato	Tokyo, Japan <i>Apr 2018–Mar 2022</i>

Professional Position

JSPS Research Fellow (PD) <i>Institute of Science Tokyo</i> Host Researcher: Kazuma Shimomoto	Apr 2026–Mar 2027
JSPS Research Fellow (DC1) <i>Tokyo Institute of Technology</i> Host Researcher: Kazuma Shimomoto	Apr 2024–Mar 2026

Research Interests

Commutative algebra in mixed characteristic (via arithmetic methods such as perfectoid rings, prismatic cohomology, and almost mathematics).

Papers and Preprints

16. **R. Ishizuka** and S. Yoshikawa, “Hodge–Tate splitting and Akizuki–Nakano vanishing in positive characteristic”, arXiv:2609.00567, 2026.
15. M. Asgharzadeh and **R. Ishizuka**, “Homological detection by perfectoid algebras”, arXiv:2607.17801, 2026.
14. **R. Ishizuka** and L. Navarro Chafloque, “On perfectoidization of finite algebras over a perfectoid ring”, arXiv:2606.12229, 2026.
13. **R. Ishizuka** and L. Navarro Chafloque, “Decomposition theorem for perfectoid rings along general ideals”, arXiv:2606.06241, 2026.
12. **R. Ishizuka** and S. Miyashita, “Trace ideals of exterior powers of the module of differentials”, arXiv:2605.00418, 2026.
11. **R. Ishizuka** and S. Yoshikawa, “A local-global correspondence for perfectoid purity”, arXiv:2604.25265, 2026.
10. **R. Ishizuka** and S. Yoshikawa, “Algebraization of absolute perfectoidization via section rings”, arXiv:2604.02682, 2026.
9. **R. Ishizuka** and S. Yoshikawa, “Derived graded modules”, arXiv:2601.19164, 2026.

¹In Oct 2024, Tokyo Institute of Technology was merged with Tokyo Medical and Dental University and reorganized into *Institute of Science Tokyo*.

8. **R. Ishizuka** and S. Yoshikawa, “*Graded perfectoid rings*”, arXiv:2511.02322, 2025.
7. **R. Ishizuka** and K. Shimomoto, “*Quasi-canonical lifting of projective varieties in positive characteristic*”, arXiv:2506.01345, 2025.
6. **R. Ishizuka**, “*A higher algebraic approach to liftings of modules over derived quotients*”, arXiv:2503.17964, 2025.
5. **R. Ishizuka**, “*Perfectoid towers generated from prisms*”, Nagoya Math. J. **261**, e17 (2026). arXiv:2409.15785, 2024.
4. **R. Ishizuka** and K. Nakazato, “*Prismatic Kunz’s theorem*”, J. Algebra **693**, 732-769 (2026). arXiv:2402.06207, 2024.
3. D. Dine and **R. Ishizuka**, “*Tilting and untilting for ideals in perfectoid rings*”, Math. Z. **307**, 66 (2024). arXiv:2308.09600, 2023.
2. **R. Ishizuka**, “*A calculation of the perfectoidization of semiperfectoid rings*”, Nagoya Math. J. **255**, 742-759 (2024). arXiv:2305.07916, 2023.
1. K. Shimomoto and **R. Ishizuka**, “*A mixed characteristic analogue of the perfection of rings and its almost Cohen-Macaulay property*”, arXiv:2303.13872, 2023.

Talks

40. Oct 2026. “*TBA*”, Kinosaki Algebraic Geometry Symposium 2026, Kinosaki International Arts Center, Japan.
39. Sep 2026. “*Local and global aspects of perfectoid pure singularities*”, The 12th Japan-Vietnam Joint Seminar on Commutative Algebra, Khanh Hoa University, Vietnam.
38. July 2026. “*Decomposition of de Rham complexes in mixed characteristic*”, Honjo Waseda Algebraic Geometry Seminar 2026, Honjo seminar house, Japan.
37. June 2026. “*Graded modules and comodules: The dual notion of modules*”, The 37th Seminar on Commutative Algebra in Japan, Chiba University, Japan.
36. Dec 2025. “*Global singularities in mixed characteristic and graded perfectoid rings*”, Mini-workshop on Arithmetic Geometry in Sendai, Tohoku University, Japan.
35. Nov 2025. “*Perfectoid towers arising from Frobenius lifts*”, The 46th Japan Symposium on Commutative Algebra, Hotel Fukuracia Osaka Bay, Japan.
34. Nov 2025. “*Perfectoid towers arising from prisms*”, Workshop on Number theory at Tsuda University 2025, Tsuda University, Japan.
33. Oct 2025. “*Absolute perfectoidization of schemes and local to global principle in mixed characteristic*”, Singularity Theory Seminar, Nihon University, Japan.
32. Sep 2025. “*The Frobenius-Witt cotangent bundle*” (Survey talk), Yatsugatake Workshop, 2025, Wellness Garden in Goddess Forest : Kobuchizawa, Japan.
31. Sep 2025. “*A version of Kunz’s theorem in mixed characteristic*”, Noda Algebraic Geometry Symposium, Tokyo University of Science, Japan.
30. July 2025. “*On the derived deformation functor of Frobenius liftings*” (Poster Session), 2025 Summer Research Institute in Algebraic Geometry., Colorado State University, USA.
29. July 2025. “*On computation of Tor modules over a derived quotient*”, The 36th Seminar on Commutative Algebra in Japan., Kyushu University, Japan.
28. June 2025. “*A unified construction of perfectoid towers by prisms*”, p -adic and Characteristic p Methods in Algebraic Geometry., EPFL, Switzerland.
27. May 2025. “*Approximation of perfectoid rings by Noetherian rings and prisms*”, MPIM Algebra Seminar., MPIM, Germany.

26. Mar 2025. "*On the vanishing of Ext and liftings of modules on derived quotients using higher algebras*", MSJ Spring Meeting 2025., Waseda University, Japan.
25. Mar 2025. "*Frobenius liftability and derived algebraic geometry*", The 29th Conference on Algebra for Young Researchers in Japan., Osaka University, Japan.
24. Mar 2025. "*Liftability results from formal moduli problems*" (including survey), Small workshop on arithmetic geometry 2025 in Hakodate., Hakodate Community Design Center, Japan.
23. Nov 2024. "*Prismatic Kunz's theorem*", The 45th Japan Symposium on Commutative Algebra., RIMS (Kyoto University), Japan.
22. Nov 2024. "*Applications of higher algebra to the lifting problem of modules*", Tokyo Commutative Algebra Seminar., Online (Zoom), Japan.
21. Oct 2024. "*F-liftability obstruction in singular varieties through derived algebraic geometry*", Singularity Theory Seminar., Nihon University, Japan.
20. Sep 2024. "*Frobenius maps on mixed characteristic rings via prismatic cohomology*" (Poster Session), L-functions and Motives in Niseko 2024., Setsu Niseko and Niseko Residents Center, Japan.
19. Sep 2024. "*A generalization of Kunz's theorem to mixed characteristic via p-adic cohomology theory*", MSJ Autumn Meeting 2024., Osaka University, Japan.
18. July 2024. "*Perfectoid spaces, tilts and untilts*" (Survey talk), Atelier de Géométrie Arithmétique 2024., RIMS (Kyoto University), Japan.
17. July 2024. "*Prismatic approach to a mixed characteristic Kunz's theorem*", The 23rd Hiroshima-Sendai Workshop on Number Theory at Sendai., Tohoku University, Japan.
16. June 2024. "*Regularity criterion of mixed characteristic rings via prismatic cohomology*", Keio Algebra Seminar., Keio University, Japan.
15. June 2024. "*Prisms and regular local rings*", The 35th Seminar on Commutative Algebra in Japan., Tokushima University, Japan.
14. Apr 2024. "*Prisms and its application to regular rings*", Saturday Seminar., Meiji University, Japan.
13. Mar 2024. "*Perfectoid ideals and its correspondence*", The 20th Mathematics Conference for Young Researchers., Hokkaido University, Japan.
12. Feb 2024. "*Mixed characteristic Kunz's theorem with prismatic theory*", The 28th Conference on Algebra for Young Researchers in Japan., Waseda University, Japan.
11. Dec 2023. "*Commutative ring theoretic approach for the perfectoidization of semiperfectoid rings*", Number Theory Seminar., Kyoto University, Japan.
10. Nov 2023. "*Ideal correspondence between a perfectoid ring and its tilt*", The 44th Japan Symposium on Commutative Algebra., Lecture Hayama, Japan.
9. Aug 2023. "*Absolute integral closure*" (Survey talk), The 18th Summer School on Commutative algebra., Tokyo Institute of Technology, Japan.
8. Aug 2023. "*On the relation between perfectoidization and p-root closure*", The 9th China-Japan-Korea International Conference on Ring and Module Theory., Incheon National University, Republic of Korea.
7. July 2023. "*On the commutative ring-theoretic structure of the perfectoidization of semiperfectoid rings*", The 22nd Hiroshima-Sendai Workshop on Number Theory at Hiroshima., Hiroshima University, Japan.
6. July 2023. "*On the application of perfectoidization to commutative algebra and its structure*", The 34th Seminar on Commutative Algebra in Japan., Kitami Institute of Technology, Japan.
5. May 2023. "*On Perfectoid(ization) and its commutative ring-theoretic properties*", Ookayama Youth Seminar in Algebra., Tokyo Institute of Technology, Japan.

4. Mar 2023. "*A mixed characteristic analogue of the perfection of rings*", The 11th Japan-Vietnam Joint Seminar on Commutative Algebra - by and for young mathematicians -, Vietnam Academy of Science and Technology, Vietnam.
3. Mar 2023. "*An explicit construction of perfectoid almost Cohen-Macaulay algebras in mixed characteristic*", MSJ Spring Meeting 2023., Chuo University, Japan.
2. Mar 2023. "*An explicit construction of perfectoid almost Cohen-Macaulay algebras*", The 19th Mathematics Conference for Young Researchers., Hokkaido University, Japan.
1. Nov 2022. "*An explicit construction of perfectoid almost Cohen-Macaulay algebras in mixed characteristic*", The 43rd Japan Symposium on Commutative Algebra., Osaka University, Japan.

Membership

- Apr 2023– . Mathematical Society in Japan (MSJ)

Languages

Japanese: Native

Mother tongue

English: Intermediate

Can read, write, and, listen but may struggle with conversation

French: Beginner

Can only read mathematical texts